Booklet No. 02237

2018

PAPER – I

Full marks – 100

Time – 1hr. 15 minutes

Signature and Name of Invigilator

1. (Signature) _____

(Name) Arijeet Saha

Time : 1¼ hours] Test Booklet Code [Y] [Maximum Marks : 100

Number of Pages in this Booklet : 16 Number of Questions in this Booklet : 60

Instructions for the Candidates

1. This paper consists of sixty (60) objective type questions, out of which the candidate would be required to answer any fifty (50) questions. In the event of the candidate attempting more than fifty questions, the first fifty questions attempted by the candidate would be evaluated.
2. At the commencement of examination, the question booklet will be given to you. In the first 5 minutes, you are required to open the booklet and compulsorily examine it as below :
   (i) To have access to the Question Booklet, tear off the paper seal on the edge of this cover page. Do not accept a booklet without sticker-seal and do not accept an open booklet.
   (ii) **Tally the number of pages and number of questions in the booklet with the information printed on the cover page. Faulty booklets due to pages/questions missing or duplicate or not in serial order or any other discrepancy should be got replaced immediately by a correct booklet from the invigilator within the period of 5 minutes. Afterwards, neither the Question Booklet will be replaced nor any extra time will be given.**
   (iii) After this verification is over, the Test Booklet Code should be entered in the OMR Sheet.
3. Each item has four alternative responses marked (A), (B), (C) and (D). You have to darken the circle as indicated below on the correct response against each item.
   Example : Ⓐ Ⓑ ● Ⓓ
   where (C) is the correct response.
4. Your responses to the items are to be indicated in the **OMR Sheet given with this Booklet, which is common for Paper I and Paper II**. If you mark at any place other than in the circle in the OMR Sheet, it will not be evaluated. **Also answer once marked shall not be cancelled or erased. If any body does it, his/her OMR sheet shall be treated as cancelled and shall not be evaluated.**
5. Read instructions given above carefully.
6. Rough Work is to be done in the end of this booklet.
7. If you write your Name, Roll Number, Phone Number or put any mark on any part of the Answer Sheet, except for the space allotted for the relevant entries, which may disclose your identity, or use abusive language or employ any other unfair means, you will render yourself liable to disqualification.
8. You have to keep the OMR sheet with you for marking the Answer of questions in Paper II. The OMR Answer sheet is to be handed over to the invigilators at the end of the examination of Paper II compulsorily and must not carry it with you outside the Examination Hall. You are, however, allowed to carry the Test Booklets for Paper I and II with the carbon copy of OMR Sheet on conclusion of examination.
9. **Use only Ball point pen.**
10. **Use of any calculator or log table etc., is prohibited.**
11. **There is no negative marks for incorrect answers.**

P.T.O.

PAPER – I

Note : There are **Sixty (60)** questions, out of which you are to answer **fifty (50)** questions only. If you answer more than **fifty (50)** questions, then only the first **fifty (50)** questions attempted will be evaluated. The questions carry **two (2)** marks each.

1. Match List I with List II and choose the correct match from the codes given below :

List I (Books)	List II (Their Authors)
1. India Wins Freedom	(i) Jawaharlal Nehru
2. Discovery of India	(ii) Bhimrao Ramji Ambedkar
3. The Accidental Prime Minister	(iii) Rabindra Nath Tagore
4. Gitanjali	(iv) Abul Kalam Azad
	(v) Humayun Kabir
	(vi) Sanjoy Baro

Codes	1	2	3	4
(A)	iv	iii	ii	v
(B)	iv	i	vi	iii
(C)	ii	vi	iii	i
(D)	v	iv	i	vi

2. Which of the following skills are needed for present day teachers to adjust effectively with the class room teaching ?

1. Knowledge of technology
2. Use of technology in teaching learning
3. Knowledge of student's needs
4. Content mastery.

(A) 1 & 3

(B) 2 & 3

(C) 2, 3 & 4

(D) 2 & 4

3. Given below are the two statements, one is Assertion (Ass) and the other is Reason (R).

Assertion (Ass) : The aerosol of the suspended particles in the atmosphere can bring about a change in climate of the region.

Reason (R) : Each of the suspended particles takes part in interactive processes established amongst them including the short waves and infra-red rays.

Select the correct answer using codes below :

Codes :

(A) Both Ass and R are true but R is not the correct explanation of Ass.

(B) Both Ass and R are true and R is the correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

4. Match the following and select the correct match from the codes given below :

a.	Tehri Dam	1.	Ganga
b.	Sardar Sarovar	2.	Beas and Satluj
c.	Indira Gandhi Canal	3.	Narmada
d.	Nagarjuna Sagar	4.	Krishna
		5.	Bhagirathi

Codes	a	b	c	d
(A)	5	3	2	4
(B)	5	3	2	1
(C)	5	3	4	2
(D)	5	3	1	2

5. Read the following two statements — Assertion (Ass) and Reason (R) and select the correct answer using codes below :

Assertion (Ass) : Robert Clive defeated Siraj-ud-Daulah in the Battle of Plassey.

Reason (R) : Clive's army was superior and followed superior military tactics.

Codes :

(A) Both Ass and R are correct and R is the correct explanation of Ass.

(B) Both Ass and R are correct but R is not a correct explanation of Ass.

(C) Ass is correct, but R is incorrect.

(D) Ass is incorrect, but R is correct.

6. An interactive tool (program) that provides an interface between the users and information available on the www is known as

(A) OPERATING SYSTEM

(B) WEB BROWSER

(C) SEARCH ENGINE

(D) URL.

7. What is 'ILLIAC' ?

(A) Super Computer

(B) Mini Computer

(C) Main Frame Computer

(D) None of the above.

8. Which of the following is not an example of an operating system ?

(A) Windows 98

(B) BSD Unix

(C) Microsoft Office XP

(D) Red Hat Linux.

9. Which of the following is the correct expanded form of INFLIBNET ?

(A) Information on library network

(B) Information and learning network

(C) Information and library network

(D) None of the above.

10. Of the following, which is the most desirable outcome of teaching in higher education ?

(A) Increase in students achievement

(B) Increase in level of independent thinking of students

(C) Higher percentage of result

(D) Increase in the number of students who opt for the subject.

11. The present growth rate of world's human population is in its

(A) lag phase

(B) exponential phase

(C) sigmoid phase

(D) none of the above.

12. A researcher uses statistical techniques in his problem to confirm

(A) whether worthwhile inferences could be drawn

(B) whether the data could be quantified

(C) whether appropriate statistical techniques are available

(D) whether analysis of data would be possible.

13. The research that applies the laws at the time of field study to draw more and more clear ideas about the problem is

(A) action research

(B) experimental research

(C) applied research

(D) survey research.

14. Which of the following is a paramount requirement of a researcher ?

(A) Scientific thinking

(B) Scientific feeling

(C) Scientific behaviour

(D) Scientific attitude.

15. The 'Times of India' came into being with the merger of

(A) 'The Bombay Times' and 'The Courier'

(B) 'The Standard' and 'The Telegraph'

(C) 'The Standard' and 'The Courier'

(D) Both (A) and (B).

16. Decoding is associated with

(A) message sender

(B) message receiver

(C) both of the above

(D) none of the above.

17. With which of the following proposition about research you do not agree ?

(A) Research improves the quality of teaching

(B) Research contributes to social progress of the country

(C) Research is a joy in itself

(D) Research leads to finding solutions.

18. The main objective of public broadcasting system i.e. Prasar Bharati is

(A) Inform, Entertainment and Education

(B) Entertain, Information and Interaction

(C) Educate, Interact and Entertain

(D) Entertainment only.

19. What is the essential step for an effective communicator ?

(A) Selection of the channel for communication

(B) Thinking about the evaluation procedure

(C) Determining the objectives of communication

(D) Identifying various mediums of communication.

20. The work of Messenger is

(A) to develop a message

(B) to encode the message

(C) to transmit the message

(D) all of the above.

21. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using codes below :

Assertion (Ass) : There is population explosion in India.

Reason (R) : India has failed to check the fertility rate.

Codes :

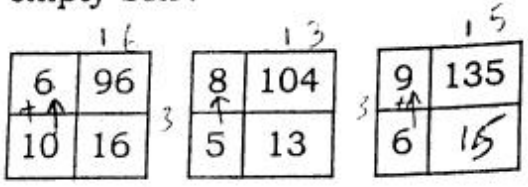
(A) Ass is true, but R is false

(B) Ass is false, but R is true

(C) Both Ass and R are true but R is not the correct explanation of Ass

(D) Both Ass and R are true and R is the correct explanation of Ass.

22. What will be the number in the empty box ?



(A) 13

(B) 15

(C) 17

(D) 19

23. A profit of Rs. 1581 is to be divided amongst three partners Pimon, Jupi and Diky in the ratio of $1/3 : 1/2 : 1/5$; what will be Diky's share ?

(A) 285

(B) 258

(C) 306

(D) 316

24. In an examination, 35% of the total students failed in Hindi, 45% failed in English and 20% in both. The percentage of those who passed in both the subjects is

(A) 10%

(B) 20%

(C) 30%

(D) 40%

25. If $X : Y = 4 : 7$, $Y : Z = 5 : 11$, then $X : Y : Z$ is

(A) 14 : 35 : 55

(B) 20 : 35 : 77

(C) 28 : 34 : 66

(D) 20 : 44 : 77

26. If 41 = 5, 23 = 5, 212 = 5, 11111 = 5 then 2592496 is equal to

(A) 9

(B) 5

(C) 3

(D) 1

27. Insert the three missing letters

AKR : DNU : : IUO : ___

(A) LXR

(B) LRX

(C) JVP

(D) KWQ

28. **Statements :**

a. All graduates are Mexicans.

b. All Mexicans are males.

Conclusions :

I. Some males are graduates.

II. All males are Mexicans.

III. Some Mexicans are graduates.

IV. All Mexicans are graduates.

On the basis of the two statements given at the top, which two conclusions are correct ?

(A) I and III

(B) I and IV

(C) II and III

(D) II and IV

29. Which of the following is the correct sequence in descending order of two persons using as mother tongue

1. Bengali 2. Marathi 3. Tamil 4. Telegu

Select the correct answer using codes given below —

(A) 1, 4, 3, 2

(B) 1, 2, 4, 3

(C) 1, 4, 2, 3

(D) 4, 2, 1, 3

30. Following are two statements — One Assertion (Ass) and the other Reason (R). Examine them and find the correct answer.

Assertion (Ass) : The urban population in India is more than that of the USA.

Reason (R) : The level of urbanisation in USA is higher than that of India.

Codes :

(A) Both Ass and R are true and R is the correct explanation of Ass.

(B) Both Ass and R are true, but R is not the correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

31. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using the codes below :

Assertion (Ass) : Proper drainage system, together with encroachment on Bharalu, Bahini and Wetlands like Silsako, Bondajan, Deepor beel have caused flood mainly in Nabin Nagar, Tarun Nagar, R. G. Barooah Road, G. S. Road and Lachit Nagar.

Reason (R) : Proper eviction drive, irrespective of political affiliation and pressure, is to be carried out by the administration to get rid of the flood.

Codes :

(A) Both Ass and R are true and R is the correct explanation of Ass.

(B) Both Ass and R are true, but R is not the correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

32. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using the codes below :

Assertion (Ass) : Areas near the equator receive rainfall throughout the year.

Reason (R) : High temperature and high humidity cause convectional rain in most afternoons near the equator.

Codes :

(A) Both Ass and R are correct and R is the correct explanation of Ass.

(B) Both Ass and R are true, but R is not the correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

33. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using the codes below :

Assertion (Ass) : The frequency of floods in the plains of Northern India including the Brahmaputra plain has increased during the last couple of decades.

Reason (R) : There has been duction in the depth of river valleys due to deposition of silt.

Codes :

(A) Both Ass and R are true and R is the correct explanation of Ass.

(B) Both Ass and R are true, but R is not a correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

34. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using the codes below :

Assertion (Ass) : The Koyana (Koyna) region of Maharashtra is likely to become more earthquake prone in near future.

Reason (R) : The Koyna dam is located on the old fault-plane which may get activated more frequently with changes in water level in Koyna reservoir.

Codes :

(A) Both Ass and R are true and R is the correct explanation of Ass.

(B) Both Ass and R are true, but R is not the correct explanation of Ass.

(C) Ass is true, but R is false.

(D) Ass is false, but R is true.

Read the following passage and answer the question Nos. **35** to **39**.

One day when I was in the fifth standard at the Rameswaram Elementary School, a new teacher came to our class. I used to wear a cap which marked me as a Muslim, and I always sat in the front row next to Ramanadha Sastry, who wore a sacred thread. The new teacher could not stomach a Hindu priest's son sitting with a Muslim boy. In accordance with our social ranking as the new teacher saw it, I was asked to go and sit on the back bench. I felt very sad, and so did Ramanadha Sastry. He looked utterly downcast as I shifted to my seat in the last row.

The image of him weeping when I shifted to the last row left a lasting impression on me.

After school, we went home and told our respective parents about the incident. Lakshmana Sastry summoned the teacher, and in our presence, told the teacher that he should not spread the poison of social inequality and communal intolerance in the minds of innocent children. He bluntly asked the teacher to either apologise or quit the school and the island. Not only did the teacher regret his behaviour, but the strong sense of conviction Lakshmana Sastry conveyed ultimately reformed this young teacher.

On the whole, the small society of Rameswaram was very rigid in terms of the segregation of different social groups. However, my science teacher Sivasubramania Iyer, though an orthodox Brahmin with a very conservative wife, was something of a rebel. He did his best to break social barriers so that people from varying backgrounds could mingle easily. He used to spend hours with me and would say, "Kalam, I want you to develop so that you are on par with the highly educated people of the big cities."

One day, he invited me to his home for a meal. His wife was horrified at the idea of a Muslim boy being invited to dine in her ritually pure kitchen. She refused to serve me in her kitchen. Sivasubramania Iyer was not perturbed, nor did he get angry with his wife, but instead, served me with his own hands and sat down beside me to eat his meal. His wife watched us from behind the kitchen door. I wondered whether she had observed any difference in the way I ate rice, drank water or cleaned the floor after the meal. When I was leaving his house, Sivasubramania Iyer invited me to join him for dinner again the next weekend. Observing my hesitation, he told me not to get upset, saying, "Once you decide to change the system, such problems have to be confronted." When I visited his house the next week, Sivasubramania Iyer's wife took me inside her kitchen and served me food with her own hands.

35. Whom do you think to be the author of the above write-up ?

(A) Maulana Abul Kalam Azad

(B) Dr APJ Abdul Kalam

(C) KA Abbas

(D) K Ataturk.

36. The meaning of the word "stomach" in the first paragraph is

(A) to eat greedily

(B) colic

(C) to tolerate

(D) abdomen.

37. The new teacher sent the author to sit on the back bench because

(A) he was not attentive

(B) he was wearing a cap

(C) he was sitting near a Hindu priest's son

(D) he was a terrible nuisance.

38. Who regretted his own behaviour ?

(A) The new teacher

(B) Ramanadha Sastry

(C) Lakshmana Sastry

(D) Sivasubramania Iyer.

39. Sivasubramania Iyer was something of a rebel because

(A) he was fighting against the British

(B) he did his best to break social barriers

(C) he was an orthodox Brahmin

(D) he married a progressive lady.

40. Burning of fossil fuels is the main source of pollution due to

(A) nitrogen oxide

(B) nitric acid

(C) nitrous oxide

(D) sulphur dioxide.

41. The slogan "A tree for each child" was coined for

(A) Environmental protection programme

(B) Clean air programme

(C) Soil conservation programme

(D) Social forestry programme.

42. Read the following two statements — Assertion (Ass) and Reason (R) and find the correct answer using the codes below :

Assertion (Ass) : The earth's outer core is liquid.

Reason (R) : The S-seismic wave cannot pass through the earth's outer core.

Codes :

(A) Both Ass and R are not correct.

(B) Both Ass and R are correct and R is the correct explanation.

(C) Ass is wrong and therefore the R has no value.

(D) Both Ass and R are correct but R is not the correct explanation of Ass.

43. Match the following river dams to the States they belong and select the correct match from the codes given below :

a. Umtru		i. Assam	
b. Lower Subansiri		ii. West Bengal	
c. Maithon		iii. Odisha	
d. Hirakud		iv. Meghalaya	
Codes	a	b	c	d
(A)	i	ii	iii	iv
(B)	iv	i	ii	iii
(C)	ii	iii	iv	i
(D)	iii	iv	i	ii

44. Match the following and select the correct match from the codes given below :

a. Barchan — i. Youth stage of rivers

b. Waterfalls — ii. Mature stage of rivers

c. Moraines — iii. Wind action deposits

d. Meanders — iv. Glacial deposits

Codes	a	b	c	d
(A)	iii	i	ii	iv
(B)	iii	i	iv	ii
(C)	iii	iv	i	ii
(D)	i	ii	iv	iii

45. Acid rains in some industrial cities are due to

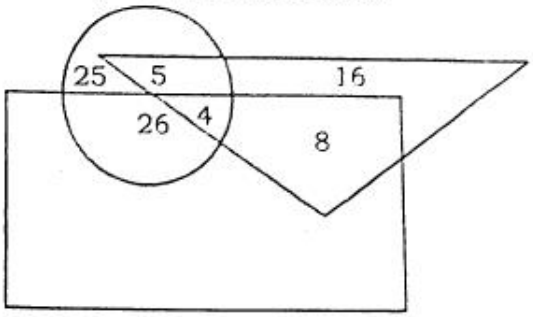
(A) excessive release of SO_2 by industries and coal gas

(B) excessive release of CO_2 by burning of fossil fuel

(C) excessive concentration of SO_2 and NO_2 in air by deforestation

(D) population explosion, deforestation and release of CO_2 by automobile exhausts.

46. In the following diagram, the rectangle represents students, the triangle represents professors and the circle represents musicians.



How many professors are both students and musicians ?

(A) 4

(B) 5

(C) 9

(D) 30

47. The following table indicates the examination scores of students in mathematics.

Scores	Number of Students
90 - 99	15
80 - 89	147
70 - 79	157
60 - 69	175
50 - 59	167
40 - 49	161
30 - 39	149
20 - 29	23
10 - 19	6

What is the percentage of students whose scores lie between 40 and 79 ?

(A) 65%

(B) 66%

(C) 67%

(D) 68%

48. Study the following incomplete table referring to number of students and average examination scores of students belonging to two sections.

Sections	Number of students	Average examination scores
Section 1	$N_1 = 43$	$M_1 = 60$
Section 2	$N_2 =$	$M_2 = 70$

If the average examination score of students of the two sections combined is 67·5, then what is the number of students in section 2 ?

(A) 55

(B) 56

(C) 57

(D) 58

49. Balen has an annual income of Rs. 2500. He spends 10 per cent on education. Then 20% of the remaining income is spent on housing. The remaining 15 per cent is deposited in the saving schemes and the rest of the income is spent on food and clothes. What percentage of income does the gentleman spend on food and clothes ?

(A) 55%

(B) 60%

(C) 61·2%

(D) 64·1%

50. Who among the following is not a Nobel Prize Winner ?

(A) Amartya Sen

(B) C. V. Raman

(C) Rabindra Nath Tagore

(D) Homi Jehangir Bhaba

51. Match List I with List II and select the correct match from the codes given below :

List I (Scientists)	List II (Field of specialisation)
1. Subrahmanyan Chandrasekhar	(i) Geology
2. Shanti Swarup Bhatnagar	(ii) Genetics
3. Srinivas Iyengar Ramanujan	(iii) Chemistry
4. M. S. Swaminathan	(iv) Statistics
	(v) Mathematics
	(vi) Astrophysics

Codes	1	2	3	4
(A)	vi	iii	v	ii
(B)	iii	v	iv	i
(C)	vi	iii	iv	i
(D)	ii	v	vi	iv

52. Match List I with List II and select the correct match from the codes given below :

List I (Abbreviations)	List II (Associated field)
1. AIU	(i) Agriculture
2. CSIR	(ii) Aeronautics
3. IARI	(iii) Education
4. ISRO	(iv) Research on science and industry
	(v) Research on space
	(vi) Robotics

Codes	1	2	3	4
(A)	iii	iv	i	v
(B)	iv	v	ii	vi
(C)	ii	v	i	iv
(D)	iii	vi	ii	v

53. Which of the following is the correct expansion of the abbreviation NCTE ?

(A) National Curriculum for Teacher Education

(B) National Council for Technical Education

(C) National Council for Teacher Education

(D) National Committee for Technical Education.

54. Match List I with List II and select the correct match from the codes given below :

List I (Commissions)	List II (Years of appointment)
1. Indian Education Commission	(i) 1905
2. Secondary Education Commission	(ii) 1917
3. University Education Commission	(iii) 1948
4. Calcutta University Commission	(iv) 1952
	(v) 1964
	(vi) 1968

Codes	1	2	3	4
(A)	v	iv	iii	ii
(B)	iv	v	ii	iii
(C)	vi	iv	iii	i
(D)	iii	vi	i	v

55. Who among the following is not a Bharat Ratna recipient ?

(A) Dr. C. V. Raman

(B) Dr. M. Visvesharaya

(C) Dr. Amartya Sen

(D) Dr. V. K. R. V. Rao.

56. The set of computer programmes that manage the hardware / software of a computer is called

(A) Computer system

(B) Operation system

(C) Operating system

(D) None of these.

57. Consider the following statements about internet facility.

I. It serves as the largest online information system

II. It provides the most dependable and latest mode of sending Email

III. Recurring expenditure on communicating Email is very high

IV. It is often misused and has triggered cyber crimes.

Which of these statements are correct ?

(A) I, II and IV

(B) I and IV

(C) II, III and IV

(D) II and III

58. The study of interaction between living organism and environment is known as

(A) autoecology

(B) ecology

(C) limacology

(D) ecosystem.

59. Match List I with List II and choose the correct match from the codes given below :

List I (Persons)	List II (Countries of their birth)
1. Charles Darwin	(i) Germany
2. Albert Einstein	(ii) France
3. Karl Marx	(iii) Austria
4. Sigmund Freud	(iv) Prussia
	(v) England
	(vi) Hungary

Codes	1	2	3	4
(A)	ii	i	v	iv
(B)	v	i	iv	iii
(C)	v	iv	i	vi
(D)	i	iii	ii	v

60. Match List I with List II and choose the correct match from the codes given below :

List I (Major Contribution)	List II (Persons)
1. Theory of Relativity	(i) Gregor Mendel
2. Periodic Table	(ii) Ivan Pavlov
3. Basic Principle of Heredity	(iii) Charles Darwin
4. Theory of Evolution	(iv) Albert Einstein
	(v) Neils Bohr
	(vi) Mendeleev

Codes	1	2	3	4
(A)	iv	vi	i	iii
(B)	v	ii	i	iii
(C)	iv	v	vi	i
(D)	i	iii	vi	ii